Rudivan Paixão Barbosa

Decision Science Analyst · Cassotis Consulting
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SUMMARY

Professional with experience in optimization and operations research, with an emphasis on mathematical modeling and algorithm development. I apply mathematical programming techniques and metaheuristics to solve planning and logistics problems.

My background includes building mathematical models for the steel industry and studying lot-sizing and scheduling, vehicle routing and facility location problems.

Technologies and tools

- Programming languages and development tools: Python, C, C++, Git
- Solvers and modeling: IBM ILOG CPLEX (Concert Technology), CONOPT, AMPL, Hexaly (formerly LocalSolver)
- Metaheuristics: GRASP, genetic algorithms, VNS, ILS, tabu search

Experience and skills

- Operations research and mathematical modeling: linear, mixed-integer, nonlinear and mixed-integer nonlinear programming models, with algorithm implementation in C/C++.
- Metaheuristics and parameter tuning: development of classical metaheuristics, with parameter calibration using iRace.
- Multiobjective optimization: application of the weighted sum method and the ε-constraint method.
- Development of mathematical programming models for lot-sizing and scheduling, vehicle routing and facility location problems.

PROFESSIONAL EXPERIENCE

Decision Science Analyst *04/2024 - Present*
Cassotis Consulting · Belo Horizonte, MG, Brazil

- Development, support and improvement of mathematical programming models for the largest steel companies in Latin America.
- Skills applied: mixed-integer optimization, nonlinear optimization, mathematical modeling, Git, AMPL, Hexaly and CONOPT.

Substitute Professor *09/2020 - 10/2021*
Federal University of Espírito Santo (UFES), Department of Applied Mathematics · São Mateus, ES, Brazil

- Courses taught: Differential and Integral Calculus, Linear Algebra, Analytic Geometry and Basic Mathematics.

EDUCATION

PhD in Computer Science (Algorithms and Mathematical Optimization) *03/2022 - In progress*
Federal University of Ouro Preto (UFOP)

- Thesis: Location-Allocation and Routing Problem for Mammography Units.
- Advisor: Marcone Jamilson Freitas Souza.

Master's in Computational Mathematics (Mathematical Optimization) *08/2017 - 08/2019*
University of São Paulo (USP)

- Dissertation: Solution approaches for the lot-sizing and scheduling problem with order acceptance.
- Advisor: Maristela Oliveira dos Santos.

Bachelor's in Mathematics (Operations Research and Optimization) *03/2012 - 01/2017*
Federal University of Espírito Santo (UFES)

- Undergraduate thesis: Optimal design of a semi-mechanized network.
- Advisor: Gilberto de Miranda Junior.

AREAS OF EXPERTISE AND SKILLS

Area/Subarea: Computer Science, Optimization.
Research topics: lot-sizing and scheduling, vehicle routing, facility location, metaheuristics.
Tools and languages: AMPL, CPLEX, Concert Technology, Hexaly, C/C++ and Git.
Methods: linear, mixed-integer, nonlinear and mixed-integer nonlinear programming; GRASP, Genetic Algorithms, VNS, ILS, LAHC, Tabu Search; parameter tuning with iRace; asymptotic analysis of algorithms; multiobjective optimization (Weighted Sum, ε-constraint).
Teaching: Differential and Integral Calculus, Linear Algebra, Analytic Geometry, Basic Mathematics.

LANGUAGES

Portuguese: native.
English: professional working proficiency.

SELECTED PUBLICATIONS

- BARBOSA, R. P. et al. Mammography Unit Location: Reconciling Maximum Coverage and Budgetary Constraints. In: 25th International Conference on Enterprise Information Systems, 2023. Link
- BARBOSA, R. P. et al. Um novo modelo para o problema de dimensionamento e sequenciamento de lotes com aceitação de pedidos [A new model for the lot-sizing and scheduling problem with order acceptance]. In: LV Brazilian Symposium on Operations Research (SBPO), 2023. In Portuguese. DOI
- BARBOSA, R. P. et al. Um modelo para o problema de dimensionamento e sequenciamento de lotes com aceitação de pedidos [A model for the lot-sizing and scheduling problem with order acceptance]. In: LI Brazilian Symposium on Operations Research (SBPO), 2019. In Portuguese. DOI
- RIBEIRO, L. T. M. et al. Dimensionamento e sequenciamento de lotes da produção: insights de uma revisão sistemática da literatura e análise bibliométrica [Production lot-sizing and scheduling: insights from a systematic literature review and bibliometric analysis]. In: ENEGEP 2024, Brazilian National Meeting of Production Engineering, 2024. In Portuguese. DOI